RECENT NEWS:

12/17/25 Five new blogs [1] [2] [3] [4] [5] on ODEs and SDEs in machine learning.

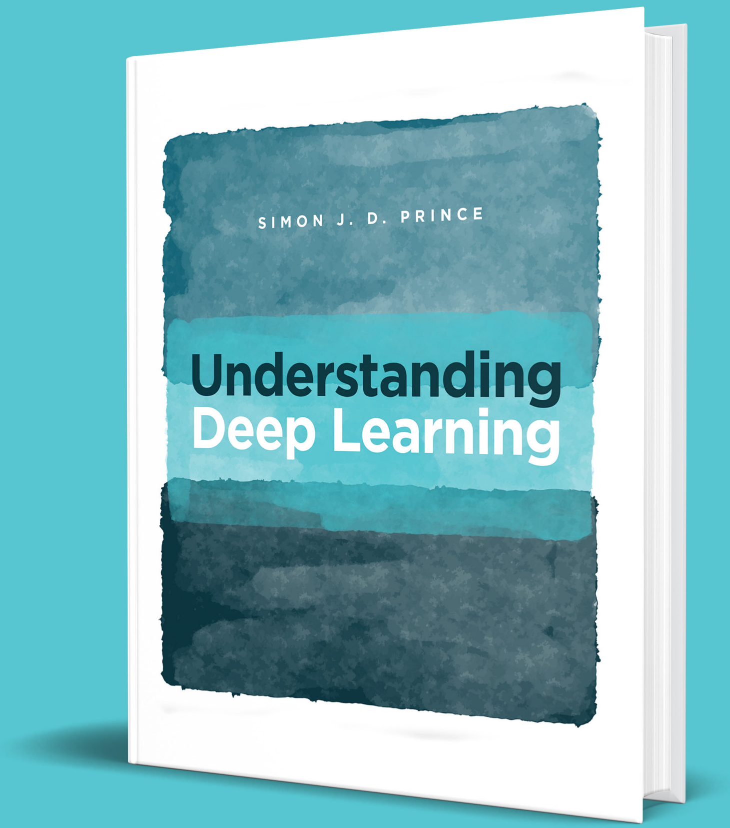
01/23/25 Added bibfile for book and LaTeX for all equations

12/17/24 Video lectures for chapters 1-12 from Tamer Elsayed of Qatar University.

12/05/24 New blog on Neural network Gaussian processes

11/14/24 New blog on Bayesian Neural Networks

08/13/24 New blog on Bayesian machine learning (function perspective)



Show more

CITATION:

```
@book{prince2023understanding,
  author = "Simon J.D. Prince",
  title = "Understanding Deep Learning",
  publisher = "The MIT Press",
  year = 2023,
  url = "http://udlbook.com"
}
```

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Download full PDF (09 Feb 2026)

Buy the book

Answers to selected questions

Errata

CODING EXERCISES

Python

notebooks covering the whole text

Sixty eight python notebook exercises with missing code to fill in based on the text

- Notebook 1.1 - Background mathematics: ipynb/colab
- Notebook 2.1 - Supervised learning: ipynb/colab
- Notebook 3.1 - Shallow networks I: ipynb/colab
- Notebook 3.2 - Shallow networks II: ipynb/colab
- Notebook 3.3 - Shallow network regions: ipynb/colab
- Notebook 3.4 - Activation functions: ipynb/colab
- Notebook 4.1 - Composing networks: ipynb/colab
- Notebook 4.2 - Clipping functions: ipynb/colab

- Notebook 11.1 - Shattered gradients: ipynb/colab
- Notebook 11.2 - Residual networks: ipynb/colab
- Notebook 11.3 - Batch normalization: ipynb/colab
- Notebook 12.1 - Self-attention: ipynb/colab
- Notebook 12.2 - Multi-head self-attention: ipynb/colab
- Notebook 12.3 - Tokenization: ipynb/colab
- Notebook 12.4 - Decoding strategies: ipynb/colab
- Notebook 13.1 - Encoding graphs: ipynb/colab
- Notebook 13.2 - Graph classification: ipynb/colab
- Notebook 13.3 - Neighborhood sampling: ipynb/colab

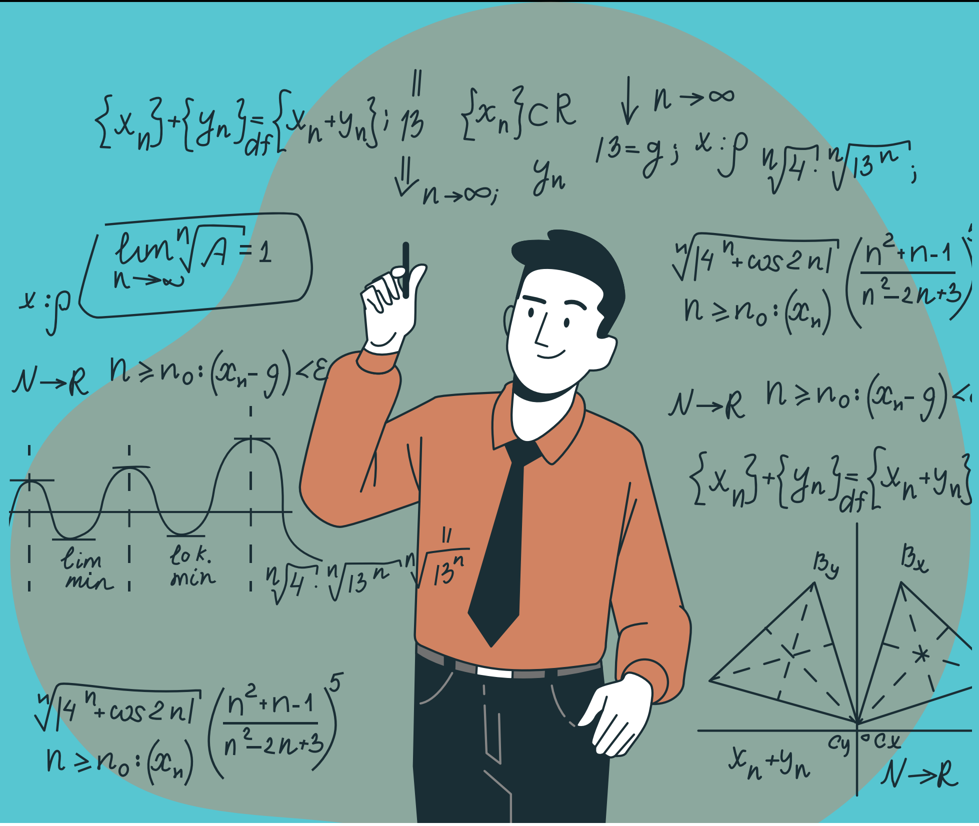
- Notebook 4.3 - Deep networks: ipynb/colab
- Notebook 5.1 - Least squares loss: ipynb/colab
- Notebook 5.2 - Binary cross-entropy loss: ipynb/colab
- Notebook 5.3 - Multiclass cross-entropy loss: ipynb/colab
- Notebook 6.1 - Line search: ipynb/colab
- Notebook 6.2 - Gradient descent: ipynb/colab
- Notebook 6.3 - Stochastic gradient descent: ipynb/colab
- Notebook 6.4 - Momentum: ipynb/colab
- Notebook 6.5 - Adam: ipynb/colab
- Notebook 7.1 - Backpropagation in toy model: ipynb/colab
- Notebook 7.2 - Backpropagation: ipynb/colab
- Notebook 7.3 - Initialization: ipynb/colab
- Notebook 8.1 - MNIST-1D performance: ipynb/colab
- Notebook 8.2 - Bias-variance trade-off: ipynb/colab
- Notebook 8.3 - Double descent: ipynb/colab
- Notebook 8.4 - High-dimensional spaces: ipynb/colab
- Notebook 9.1 - L2 regularization: ipynb/colab
- Notebook 9.2 - Implicit regularization: ipynb/colab
- Notebook 9.3 - Ensembling: ipynb/colab
- Notebook 9.4 - Bayesian approach: ipynb/colab
- Notebook 13.4 - Graph attention: ipynb/colab
- Notebook 15.1 - GAN toy example: ipynb/colab
- Notebook 15.2 - Wasserstein distance: ipynb/colab
- Notebook 16.1 - 1D normalizing flows: ipynb/colab
- Notebook 16.2 - Autoregressive flows: ipynb/colab
- Notebook 16.3 - Contraction mappings: ipynb/colab
- Notebook 17.1 - Latent variable models: ipynb/colab
- Notebook 17.2 - Reparameterization trick: ipynb/colab
- Notebook 17.3 - Importance sampling: ipynb/colab
- Notebook 18.1 - Diffusion encoder: ipynb/colab
- Notebook 18.2 - 1D diffusion model: ipynb/colab
- Notebook 18.3 - Reparameterized model: ipynb/colab
- Notebook 18.4 - Families of diffusion models: ipynb/colab
- Notebook 19.1 - Markov decision processes: ipynb/colab
- Notebook 19.2 - Dynamic programming: ipynb/colab
- Notebook 19.3 - Monte-Carlo methods: ipynb/colab

- Notebook 9.5 - Augmentation: ipynb/colab
- Notebook 10.1 - 1D convolution: ipynb/colab
- Notebook 10.2 - Convolution for MNIST-1D: ipynb/colab
- Notebook 10.3 - 2D convolution: ipynb/colab
- Notebook 10.4 - Downsampling & upsampling: ipynb/colab
- Notebook 10.5 - Convolution for MNIST: ipynb/colab
- Notebook 19.4 - Temporal difference methods: ipynb/colab
- Notebook 19.5 - Control variates: ipynb/colab
- Notebook 20.1 - Random data: ipynb/colab
- Notebook 20.2 - Full-batch gradient descent: ipynb/colab
- Notebook 20.3 - Lottery tickets: ipynb/colab
- Notebook 20.4 - Adversarial attacks: ipynb/colab
- Notebook 21.1 - Bias mitigation: ipynb/colab
- Notebook 21.2 - Explainability: ipynb/colab

INSTRUCTORS

Resources for instructors

All the figures in vector and image formats, full slides for first twelve chapters, instructor answer booklet



REGISTER

Register with MIT Press for answer booklet.

INTERACTIVE FIGURES

Interactive figures to illustrate ideas in class

FULL SLIDES

Slides for 20 lecture undergraduate deep learning course:

1. Introduction PPTX
2. Supervised Learning PPTX
3. Shallow Neural Networks PPTX
4. Deep Neural Networks PPTX
5. Loss Functions PPTX
6. Fitting Models PPTX
7. Computing Gradients PPTX
8. Initialization PPTX
9. Performance PPTX
10. Regularization PPTX
11. Convolutional Networks PPTX
12. Image Generation PPTX
13. Transformers and LLMs PPTX

FIGURES

1. Introduction: PDF / SVG / PPTX
2. Supervised learning: PDF / SVG / PPTX
3. Shallow neural networks: PDF / SVG / PPTX
4. Deep neural networks: PDF / SVG / PPTX
5. Loss functions: PDF / SVG / PPTX
6. Training models: PDF / SVG / PPTX
7. Gradients and initialization: PDF / SVG / PPTX
8. Measuring performance: PDF / SVG / PPTX
9. Regularization: PDF / SVG / PPTX
10. Convolutional networks: PDF / SVG / PPTX
11. Residual networks: PDF / SVG / PPTX
12. Transformers: PDF / SVG / PPTX
13. Graph neural networks: PDF / SVG / PPTX
14. Unsupervised learning: PDF / SVG / PPTX
15. GANs: PDF / SVG / PPTX
16. Normalizing flows: PDF / SVG / PPTX
17. Variational autoencoders: PDF / SVG / PPTX
18. Diffusion models: PDF / SVG / PPTX
19. Deep reinforcement learning: PDF / SVG / PPTX
20. Why does deep learning work?: PDF / SVG / PPTX
21. Deep learning and ethics: PDF / SVG / PPTX
22. Appendices: PDF / SVG / PPTX

LATEX FOR EQUATIONS

A working Latex file containing all of the equations

Instructions for editing equations in figures.

LATEX BIBFILE

The bibfile containing all of the references

MEDIA

Reviews, videos, podcasts, interviews

Various resources connected to the book

REVIEWS

Machine learning street talk podcast

- Nature Machine Intelligence review by Ge Wang
- Amazon reviews
- Goodreads reviews
- Book review by Vishal V.
- Book review by Nidhin Chandrasekharan
- Book review by Justin Skycak

INTERVIEWS

- Borealis AI interview
- Shepherd ML book recommendations

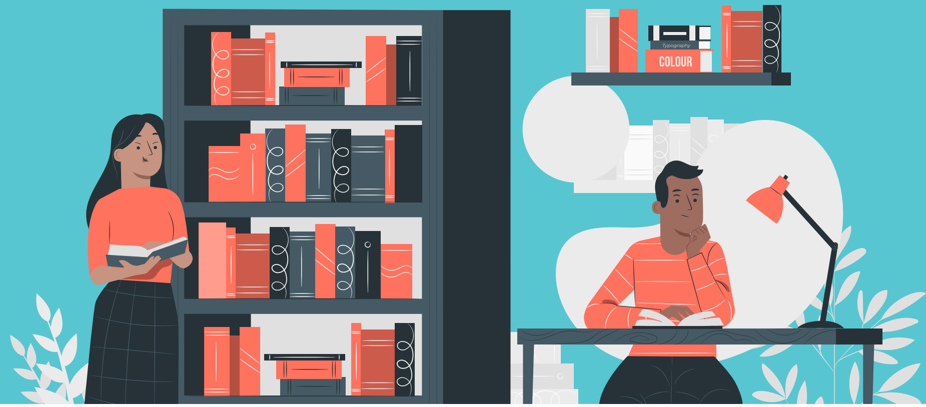
VIDEO LECTURES

- Video lectures for chapters 1-12 from Tamer Elsayed

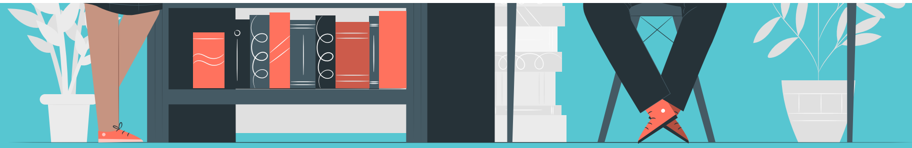
MORE

Further reading

Other articles, blogs, and books that I have written. Most in a similar style and using the same notation as Understanding Deep



Learning.



COMPUTER VISION BOOK

- Computer vision: models, learning, and inference
  - 2012 book published with CUP
  - Focused on probabilistic models
  - Pre-"deep learning"
  - Lots of ML content
  - Individual chapters available below

TRANSFORMERS & LLMS

- Intro to LLMs
  - What is an LLM?
  - Pretraining
  - Instruction fine-tuning
  - Reinforcement learning from human feedback
  - Notable LLMs
  - LLMs without training from scratch
- Transformers I
  - Dot-Product self-attention
  - Scaled dot-product self-attention
  - Position encoding

ML THEORY

- Gradient flow
  - Gradient flow
  - Evolution of residual
  - Evolution of parameters
  - Evolution of model predictions
  - Evolution of prediction covariance
- Neural tangent kernel
  - Infinite width neural networks
  - Training dynamics
  - Empirical NTK for shallow network
  - Analytical NTK for shallow network
  - Empirical NTK for deep network
  - Analytical NTK for deep network
- NTK applications
  - Trainability
  - Convergence bounds
  - Evolution of parameters
  - Evolution of predictions
  - NTK Gaussian processes
  - NTK and generalizability
- Bayesian ML I

- Multiple heads
- Transformer block
- Encoders
- Decoders
- Encoder-Decoders

- Transformers II
  - Sinusoidal position embeddings
  - Learned position embeddings
  - Relatives vs. absolute position embeddings
  - Extending transformers to longer sequences
  - Reducing attention matrix size
  - Making attention matrix sparse
  - Kernelizing attention computation
  - Attention as an RNN
  - Attention as a hypernetwork
  - Attention as a routing network
  - Attention and graphs
  - Attention and convolutions
  - Attention and gating
  - Attention and memory retrieval
- Transformers III
  - Tricks for training transformers
  - Why are these tricks required?
  - Removing layer normalization
  - Balancing residual dependencies
  - Reducing optimizer variance
  - How to train deeper transformers on small

- Maximum likelihood
- Maximum a posteriori
- The Bayesian approach
- Example: 1D linear regression
- Practical concerns

- Bayesian ML II
  - Function space
  - Gaussian processes
  - Inference
  - Non-linear regression
  - Kernels and the kernel trick
- Bayesian neural networks
  - Sampling vs. variational approximation
  - MCMC methods
  - SWAG and MultiSWAG
  - Bayes by backprop
  - Monte Carlo dropout
- Neural network Gaussian processes
  - Shallow networks as GPs
  - Neural network Gaussian processes
  - NNGP Kernel
  - Kernel regression
  - Network stability

UNSUPERVISED LEARNING

- Modeling complex data densities

datasets

- Training and fine-tuning LLMs
  - Large language models
  - Pretraining
  - Supervised fine tuning
  - Reinforcement learning from human feedback
  - Direct preference optimization
- Speeding up inference in LLMs
  - Problems with transformers
  - Attention-free transformers
  - Complexity
  - RWKV
  - Linear transformers and performers
  - Retentive network

MATH FOR MACHINE LEARNING

- Linear algebra
  - Vectors and matrices
  - Determinant and trace
  - Orthogonal matrices
  - Null space
  - Linear transformations
  - Singular value decomposition
  - Least squares problems
  - Principal direction problems
  - Inversion of block matrices
  - Hidden variables
  - Expectation maximization
  - Mixture of Gaussians
  - The t-distribution
  - Factor analysis
  - The EM algorithm in detail
- Variational autoencoders
  - Non-linear latent variable models
  - Evidence lower bound (ELBO)
  - ELBO properties
  - Variational approximation
  - The variational autoencoder
  - Reparameterization trick
- Normalizing flows: introduction and review
  - Normalizing flows
  - Elementwise and linear flows
  - Planar and radial flows
  - Coupling and auto-regressive flows
  - Coupling functions
  - Residual flows
  - Infinitesimal (continuous) flows
  - Datasets and performance

GRAPHICAL MODELS

- Graphical models
  - Conditional independence

- Inversion of block matrices
- Schur complement identity
- Sherman-Morrison-Woodbury
- Matrix determinant lemma

- Introduction to probability
  - Random variables
  - Joint probability
  - Marginal probability
  - Conditional probability
  - Bayes' rule
  - Independence
  - Expectation

- Probability distributions
  - Bernouilli distribution
  - Beta distribution
  - Categorical distribution
  - Dirichlet distribution
  - Univariate normal distribution
  - Normal inverse-scaled gamma distribution
  - Multivariate normal distribution
  - Normal inverse Wishart distribution
  - Conjugacy

- Fitting probability distributions
  - Maximum likelihood
  - Maximum a posteriori
  - Bayesian approach
  - Example: fitting normal

- Directed graphical models
- Undirected graphical models
- Inference in graphical models
- Sampling in graphical models
- Learning in graphical models

- Models for chains and trees
  - Hidden Markov models
  - Viterbi algorithm
  - Forward-backward algorithm
  - Belief propagation
  - Sum product algorithm
  - Extension to trees
  - Graphs with loops

- Models for grids
  - Markov random fields
  - MAP inference in binary pairwise MRFs
  - Graph cuts
  - Multi-label pairwise MRFs
  - Alpha-expansion algorithm
  - Conditional random fields

MACHINE LEARNING

- Learning and inference
  - Discriminative models
  - Generative models
  - Example: regression
  - Example: classification

  - Example: fitting categorical

- The normal distribution
  - Types of covariance matrix
  - Decomposition of covariance
  - Linear transformations
  - Marginal distributions
  - Conditional distributions
  - Product of two normals
  - Change of variable formula

OPTIMIZATION

- Gradient-based optimization
  - Convexity
  - Steepest descent
  - Newton's method
  - Gauss-Newton method
  - Line search
  - Reparameterization

  - Kernel choice
  - Learning GP parameters
  - Tips, tricks, and limitations
  - Beta-Bernoulli bandit
  - Random forests for BO

  - Example: classification

- Regression models
  - Linear regression
  - Bayesian linear regression
  - Non-linear regression
  - Bayesian non-linear regression
  - The kernel trick
  - Gaussian process regression
  - Sparse linear regression
  - Relevance vector regression

- Classification models
  - Logistic regression
  - Bayesian logistic regression
  - Non-linear logistic regression
  - Gaussian process classification
  - Relevance vector classification
  - Incremental fitting: boosting and trees
  - Multi-class logistic regression

- Few-shot learning and meta-learning I

  - Relation networks

- Few-shot learning and meta-learning II
  - MAML & Reptile
  - LSTM based meta-learning

- Random forests for BO
- Tree-Parzen estimators

* SAT Solvers I
  - Boolean logic and satisfiability
  - Conjunctive normal form
  - The Tseitin transformation
  - SAT and related problems
  - SAT constructions
  - Graph coloring and scheduling
  - Fitting binary neural networks
  - Fitting decision trees
* SAT Solvers II
  - Conditioning
  - Resolution
  - Solving 2-SAT by unit propagation
  - Directional resolution
  - SAT as binary search
  - DPLL
  - Conflict driven clause learning
* SAT Solvers III
  - Satisfiability vs. problem size
  - Factor graph representation
  - Max product / sum product for SAT
  - Survey propagation
  - SAT with non-binary variables
  - SMT solvers

TEMPORAL MODELS

- Reinforcement learning based approaches
- Memory augmented neural networks
- SNAIL
- Generative models
- Data augmentation approaches

NATURAL LANGUAGE PROCESSING

* Neural natural language generation I
  - Encoder-decoder architecture
  - Maximum-likelihood training
  - Greedy search
  - Beam search
  - Diverse beam search
  - Top-k sampling
  - Nucleus sampling
* Neural natural language generation II
  - Fine-tuning with reinforcement learning
  - Training from scratch with RL
  - RL vs. structured prediction
  - Minimum risk training
  - Scheduled sampling
  - Beam search optimization
  - SeaRNN
  - Reward-augmented maximum likelihood
* Parsing I
  - Parse trees
  - Context-free grammars

TEMPORAL MODELS

- Temporal models
  - Kalman filter
  - Smoothing
  - Extended Kalman filter
  - Unscented Kalman filter
  - Particle filtering

COMPUTER VISION

- Image Processing
  - Whitening
  - Histogram equalization
  - Filtering
  - Edges and corners
  - Dimensionality reduction
- Pinhole camera
  - Pinhole camera model
  - Radial distortion
  - Homogeneous coordinates
  - Learning extrinsic parameters
  - Learning intrinsic parameters
  - Inferring three-dimensional world points
- Geometric transformations
  - Euclidean, similarity, affine, projective transformations
  - Fitting transformation models
  - Inference in transformation models

  - Context-free grammars
  - Chomsky normal form
  - CYK recognition algorithm
  - Worked example
- Parsing II
  - Weighted context-free grammars
  - Semirings
  - Inside algorithm
  - Inside weights
  - Weighted parsing
- Parsing III
  - Probabilistic context-free grammars
  - Parameter estimation (supervised)
  - Parameter estimation (unsupervised)
  - Viterbi training
  - Expectation maximization
  - Outside from inside
  - Interpretation of outside weights
- XLNet
  - Language modeling
  - XLNet training objective
  - Permutations
  - Attention mask
  - Two stream self-attention

RESPONSIBLE AI

- Bias and fairness

- Three geometric problems for planes
- Transformations between images
- Robust learning of transformations

- Multiple cameras
  - Two view geometry
  - The essential matrix
  - The fundamental matrix
  - Two-view reconstruction pipeline
  - Rectification
  - Multiview reconstruction

REINFORCEMENT LEARNING

- Transformers in RL
  - Challenges in RL
  - Advantages of transformers for RL
  - Representation learning
  - Transition function learning
  - Reward learning
  - Policy learning
  - Training strategy
  - Interpretability
  - Applications

ODES AND SDES IN MACHINE LEARNING

- ODEs and SDEs in machine learning
  - ODEs

- Sources of bias
- Demographic Parity
- Equality of odds
- Equality of opportunity
- Individual fairness
- Bias mitigation

- Explainability I
  - Taxonomy of XAI approaches
  - Local post-hoc explanations
  - Individual conditional explanation
  - Counterfactual explanations
  - LIME & Anchors
  - Shapley additive explanations & SHAP

- Explainability II
  - Global feature importance
  - Partial dependence & ICE plots
  - Accumulated local effects
  - Aggregate SHAP values
  - Prototypes & criticisms
  - Surrogate / proxy models
  - Inherently interpretable models

- Differential privacy I
  - Early approaches to privacy
  - Fundamental law of information recovery
  - Differential privacy
  - Properties of differential privacy